急 救 时 刻 蓝 仕 威 克

蓝仕威克

Service Emergency Rescue

Product Instructions

MCPR 100 Series CPR

广州蓝仕威克医疗科技有限公司

Guangzhou Landswick Medical Technologies Limited

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Part I: Manufacturer information

Manufacturer：Guangzhou Landswick Medical Technologies Ltd.

ADD:NO.1302/13F, Block C2, NO.182 Science Avenue, Guangzhou Science City, Guangzhou Hi-tech Industrial Development Zone, Guangzhou City, China

Tel：020-28065380

Fax:020-28065390

Post code:510663

网 址：www.landswick.com

Part II: Product Description

MCPR 100 series CPR using pneumatic power control mode, a compressed air source is powered by the chip through the circuit control, chest (heart) and pressing interval cross ventilation. This can happen to most unexpected cardiac arrest patient to provide adequate life-sustaining oxygen and blood circulation, to the role of cardiopulmonary resuscitation. Implementation heart altar outside the clinical medical units and hospital emergency hospital arrest patients, but also applies to natural disasters caused casualties and battlefield first aid emergency ambulance.

Equipment to be used in conjunction with an oxygen source can be connected to the hospital central oxygen supply system, you can also use the connection to the mobile oxygen bottles.

Part III: product labels, graphics, symbols

Product Name / Description	Labels,graphics	Model
Nameplate Manufacturer information and product information		Common
In line with GB 9706.1-2007 standard for a high degree of protection against electric shock		Common
Prevent splashing water on each device in the direction of the housing harmful effects	IPX4	Common
Locking and unlocking the keypad to prevent accidental press, mistakenly hit led to Cheng Order situation confusion occurs		Common
Power on/off		Common
Pressing button Adjustable continuous pressing CCV, 30: 2,15: 2 three kinds of pressing mode		Common
Ventilation button Adjustable 1 second, 1.5 seconds, 2 seconds		Common
press frequency button Adjustable 100 beats / min, 110 beats / min, 120 beats / min		Common
Battery power display The battery is full , three lights show green,When the power is reducing the		Common

first light turns red		
Ventilation flow Green: ventilation 1S corresponding ventilation 0 ~ 750ml Yellow: ventilation time 1.5S corresponds to 0 ~ 1125ml Red:ventilation 2S corresponding to 0 ~ 1500ml		MCPR 100C
Ventilation flow identification ventilation time 1S corresponds to 0 ~ 750ml ventilation time 1.5S corresponds to 0 ~ 1125ml ventilation time 2S corresponds to 0 ~ 1500ml		MCPR 100B
O2 is identified as an oxygen gas source input connector		Common

Part IV:Product technical parameters

The product normal working conditions

Working temperature (℃)	Relative humidity	Scope of oxygen pressure (Kpa)	Supply voltage (V)	Power (W)
-40-55	≤93%	345-600	100-240	50

Basic parameters of products

Weight (kg)	Volume(cm) Length * width * height	Waterp roof level	Prevent electric shock	Battery last Time (h)	The battery Time (h)
≤6.5	≤60*46*7	IPX4	BF	≥8	≤4

Product performance

（1）Press the ventilation、ventilation、Compression depth、 press frequency、ventilation volume

<table>
<tr><th>Model</th><th>Press the ventilation</th><th>ventilation time（s）</th><th>Compression depth（cm）</th><th>press frequency（beats / min）</th><th>ventilation volume（mL）</th></tr>
<tr><td>MCPR 100A</td><td>Adjustable continuous pressing CCV, 30: 2,15: 2，no automatic ventilation</td><td>—</td><td rowspan="3">0-7</td><td rowspan="3">100±5%
110±5%
120±5%</td><td>—</td></tr>
<tr><td>MCPR 100B</td><td rowspan="2">Adjustable continuous pressing CCV, 30: 2,15: 2</td><td rowspan="2">1、1.5、2</td><td rowspan="2">0-1500±20%</td></tr>
<tr><td>MCPR 100C</td></tr>
</table>

（2）The pressing pressure is adjustable output when the input air pressure is 345kPa (50psi) when pressing the output pressure should not be less than 67.95Kg (150 lbs.); When the input air pressure is 600kPa (90psi), the output pressure should be pressed not more than 110Kg (245 pounds).

（3）Within a predetermined range of oxygen input, press and release time ratio of 1: 1 (50%: 50%), tolerance ± 5%.

（4）With input gas pressure is too large automatic pressure relief protection function, when the input oxygen pressure ≥0.7MPa, pressure relief valve opens.

（5）Airway pressure ≥50cmH2O, pressure indicating device automatically send

tweets.

（6）Power with oxygen tube on the oxygen pressure gauge that indicates oxygen pressure, oxygen pressure gauge accuracy in the range of ± 3%.

（7）Battery operating time ≥8 hours, charging time ≤4 hours; battery is full grid three grid, the green indicator lights; low battery of a cell, indicator lights red, indicating charging.

（8）Smart button panel with a safety lock function

（9）Having ventilation voice prompts, that is, when prompted beep when pressed to the inverse of the number 5

Part V: contraindications, scope and precautions

Contraindications

Crushed chest, pneumothorax, significant penetrating chest trauma, rib or sternum fracture, rupture of the heart patients, pregnant women, children and infants prohibited.

Notes and warnings

This product is for use by qualified medical personnel.

（1）Compression depth and ventilation volume should not exceed the desired value, so as to avoid unnecessary damage to the thoracic and soft tissue

（2）must ensure that the hosts firmly fixed patient with cardiopulmonary resuscitation machine, if it is loose, it may be in transit, the patient and the relative displacement piston cylinder, causing damage.

（3）When used in conjunction with other devices (defibrillators, pacemakers), except to note that CPR considerations, but also pay attention to considerations other devices.

（4）As a result of oxygen used in the process, we must take the usual fire prevention measures.

（5）If the work process uses lithium battery-powered, it must pay attention to the battery indicator; if power display "red light", should connect the charger to ensure continuous supply power.

（6）Recommendation fully read and understand the Operator's Manual before operating the CPR. In order to maintain efficient operation of the instrument, the operator should always be trained and repeated practice.

（7）Before CPR, medical patient should be artificial CPR.

（8）Health care workers should pay close attention to the work of the state apparatus and the actual condition of the patient, if necessary, the instrument parameter adjustment or recovery and timely manner.

（9）Battery Caution

The battery charge after 2-3 cycles, you can achieve the best capacity.

Prohibited to discharge or charge the battery positive and negative terminals short-circuit the positive and negative terminals, to avoid danger.

Avoid battery water moisture.

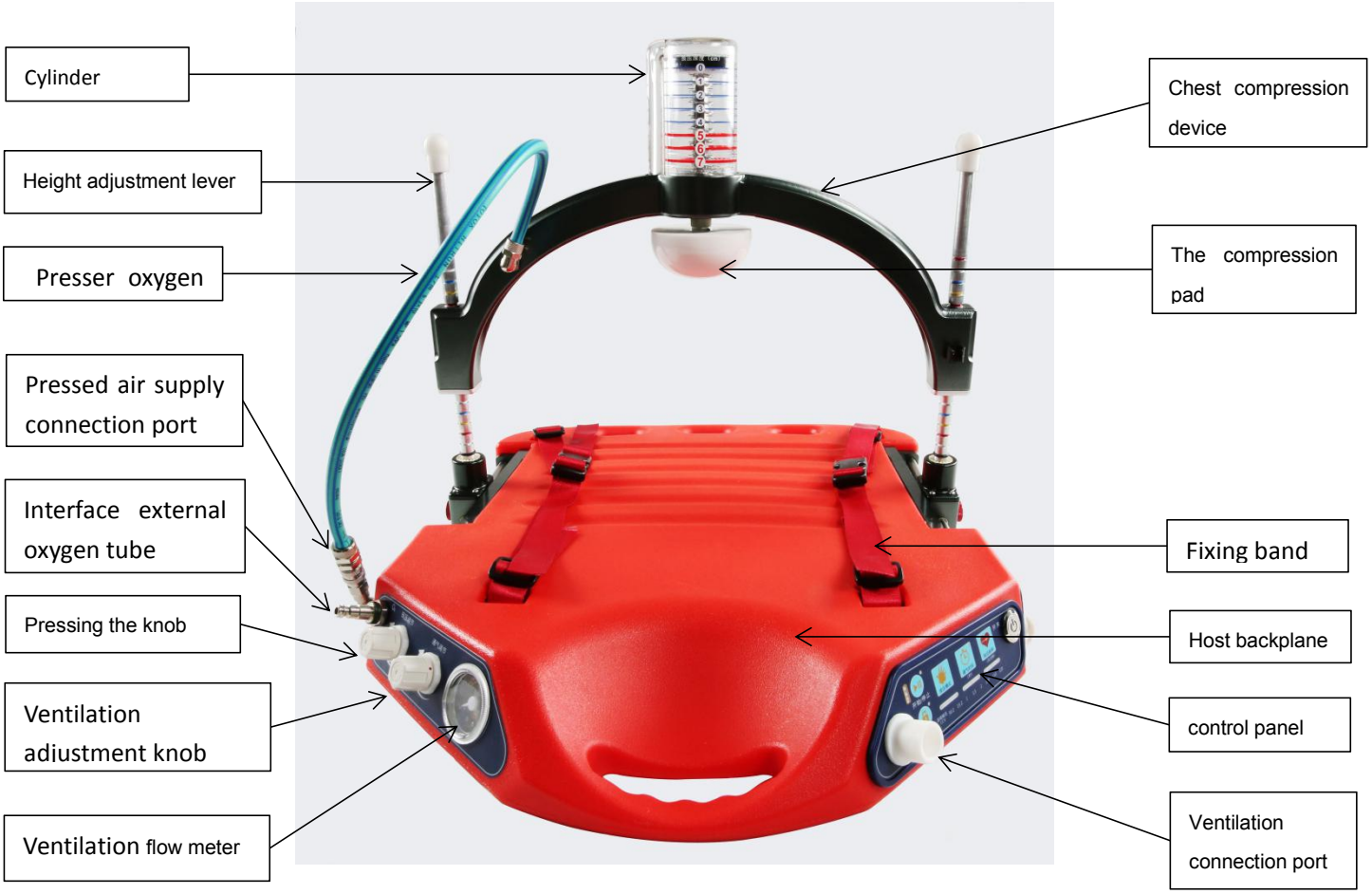
Surroundings should be kept charging dry, ventilated, non-fire source.

The charging process such as unusual circumstances, such as bulging deformation of the battery, high temperature hot, should immediately stop charging.

Part VI: Product Composition

CPR by the recovery back, the host, chest compressions and other components

See below（MCPR 100C）



Power Adapter

battery

Oxygen pipeline

Part VII: Before using the security check

1、before using CPR, check the lithium battery. CPR open the power switch, power button panel

Electricity as three are on the "green light" indicating a full charge; as bright "red light", indicating low battery, the need for timely charge

Full electrical power or replace batteries.

2、Before use, inspect the regulator knob is pressed, the ventilation regulator knob is in the "zero" state.

3、Such as the use of oxygen bottles as a mobile air source, prior to use to determine the amount of oxygen gas bottles.

Part VIII: Installation, usage and icon

Placement method

（1）Raise patient's head and shoulders , stuffed CPR from the patient back

（2）depends on the patient's head back recesses (stretch so that the patient's head, and thus help open the respiratory tract), and is fixed with a fixing tape to the patient.

Press the assembly

（1）Remove the chest compression device, the height adjustment rod is inserted in the corresponding backplane recovery slider hole, I heard a click sound explanation height adjustment lever and backplane connectors already in place and locked.

（2）The chest compression device of the trachea and the backplane connector corresponding jacks.

（3）hold the card adjustment, adjust down chest compression device, the pressing pad close to the patient at the center of the sternum 1/2 (can be adjusted by the position of the slider left and right sides of the back, locked in place after adjustment fixed screws).

Air input

（1）Oxygen pipeline connected to the air port; Suggest to adjust air pressure to 0.5 MPa.

（2）connect on the air input and backplane identification of O2

Battery Installation
Insert the battery into the battery compartment.

Start

（1）Press the power switch, turn on the power.

（2）Press 1 second Control Panel "Start / Pause" button, the device starts. Device initial operating mode as shown: Press the Mode 30: 2; ventilation 1S; pressing frequency of 100 beats / min.

（3）Adjusting the depression adjustment knob clockwise to gradually increase the pressing force, the pressing depth of the observation scale display of the cylinder, pressing until a suitable depth.

（4）Adjusting the ventilation adjustment knob is rotated in a clockwise direction, according to the display scale ventilation (MCPR 100B) or ventilation flow meter (MCPR 100C), the cyclone will pass to the appropriate scale ventilation.

Breathing masks link
A respiratory mask assembly connection

The respiratory mask assembly of oxygen in the vent end of the input interface, one end fixed to the patient's face mask, then the mask straps fixed with the patient.

Select mode

（1）pressing Mode: Continuous press / CCV mode, 30: 2 mode, 15: 2 mode

（2）ventilation time: 1S, 1.5S, 2S (applicable to MCPR 100B, MCPR 100C)

（3）pressing frequency: 100 beats / min, 110 beats / min, 120 beats / min

（4）After the parameters are set, press the "Lock / Unlock" button, lock button keypad.

stop using the CPR

（1）to suspend or stop using the CPR, and then click Close to press the knob, ventilation adjustment knob, close the "Start / Pause" button to turn off the power switch, remove the pressing system and ventilation system, disconnect the oxygen supply system.

（2）remove the battery from the machine.

（3）After use, clean and properly placed:

1. Check all the pressure knob and power switch is turned off confirmation.

2. Cleaning member in contact with the patient, such as the entire back surface of the chest compression device contact area, respirators may be contaminated with other components.

Section 9: Troubleshooting

Potential problems/ hazards	reason	Solution
Device does not work inside discouraged sound	Oxygen tube pressure above 0.7MPa or 7KG	Check oxygen intake pressure
Equipment compression depth less than 5cm, appear Respiratory than confusion	1.Check whether the pressure of the oxygen tube in 0.4 MPa-0.62MPa within range 2.Check whether pressing the knob to adjust the maximum	1.Check oxygen intake pressure 2.Call after service.
Meet the following conditions, the device is not working: 1. Enter the oxygen pressure in the range of 0.4MPa-0.62MPa 2.Under normal device connection and start state	1. Check whether there are pressing discouraged and depressed internal plunger for crack 2. Press the adjustment knob Fault 3. Control circuit failure	Call after service
Pressing frequency is less than 100 beats per minute	Logic control panel failure	Call after service
Device is turned on, but the normal regulatory can not control panel functions	1.Control panel Fault 2. Power management board failure	Call after service
Adjustment knob is pressed, the pressing device Not often pressed	1.Check equipment pressure apparatus for frustrated 2.Press knob Fault 3.Control panel Fault 4.Electromagnetic valve failure	Call after service
Only use alternating current (ac), the equipment can't work normally	1. power adapter failure 2.Power management board failure 3.Logic control board failure	Call after service
Use battery and ac connection , the equipment can't work normally	1.Power management board failure 2.Logic control board failure	Call after service
Use only batteries, equipment can't work normally	1.Damage of batteries or battery 2.Power management board failure	1.Connect set instruments recharged 2.Call after service
The power switch cannot be started	power switch failure	Call after service
Component function by adjusting the parameters of the various functions properly	1.Control panel Fault 2.control panel has been locked	1.Call after service 2.Checking whether the lock button is locked
Equipment start normal, but pressure and ventilation function does not work	1.Control panel Fault 2.Air is not open	1.Call after service 2.Checking whether the air switch is on
Equipment in the process of the work has a gas leak voice	Line seal is lax or inside the equipment aging	Call after service

Section 10: Cleaning and maintenance

Products need to regularly check the performance of the various components, it is recommended basic monthly performance checks to ensure that CPR is in normal use. After the sterilization of medical products use alcohol to clean the equipment required and the pressing external housing.

After use, the battery should be removed from the device.

Battery storage should be in about 50% of the battery capacity, storage time should not exceed 30 days. During storage charging about once every 30 days, about 2 hours per charge.

After cardiopulmonary resuscitation, should clean the parts in contact with the patient, such as increased backplane external shoulder, chest compression device contact area, respirators and other potentially contaminated parts.

Section 11　package、Transportation、Storage

1 .package

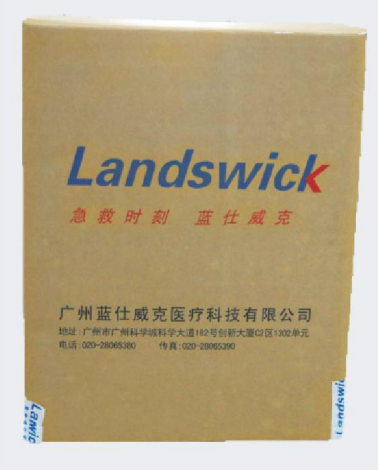
Put the entire products into the portable bag, the whole package into cartons.

2 .Transportation

Allows the use of transport in general, but it should be prevented during transportation severe shock, vibration, and rain and snow shower splash.

3. Storage

CPR packaging should be stored at ambient temperature -40 ℃ ～55 ℃, relative humidity less than 93%, non-corrosive gases, well-ventilated room.



Section 12 Quality assurance

The free warranty is one year.for quality problem Due to operational problems caused by an error is not within the scope of free maintenance.

To ensure service or repair, if the product is abnormal, the buyer must contact Landswick maintenance department. The maintenance department will try to communicate by telephone to resolve the problem first . If confirmed the product needs to be repaired, the product will be returned to Landswick company ,transportation costs pre-paid by the buyer.

Without landswick authorization, the machine must be sent back with the original packaging together .Once the machine returns to the factory, and after checking .Landswick will determine whether under warranty. If in warranty period, The Landswick company will give free repair or replacement of substandard products and send back the machine with paying shipping cost.

If the product not covered under warranty, the customer will decide repair or not, maintenance costs and transportation costs will be paid by the customer.

The above warranty conditions do not apply the following defects or damage caused by:

1. improper use or abuse
2.neglect, fire, flood, loss, theft
3.normal wear and tear
4. improper or inadequate maintenance
5. unauthorized modification or repair
6. the use of unauthorized accessories or consumables
7.used or stored method parameters out of Landswick identified

Do not do the following quality assurance:

1.any component or system open product or service by an unauthorized service personnel
2.any component or system use of the incompatible accessories or parts
3.Landswick advised returned for repairing, but the buyer refused returned for service or repair
4.the customer is not in accordance with the product specification requirements and maintenance products
5.the product is not used in accordance with the intended use of the product instructions provided